# Narration Script

## MAP-A rationale video v5 · how MCS comes into play · version without narration

Length 1:56 · 1920 × 1080 · 30 fps · subtitles burned in · 243 words in four scenes, then a 10-second title card with no narration. Read each line over **MAP-A_Rationale_Video_v5_MCS_No_Narration.mp4**. Each line starts when its subtitle appears, and every visual cue is tied to a word, so match your pace to the subtitles.

| How to record | Say it this way |
|---|---|
| 1. Play the video and start speaking when each subtitle appears (**Start**).<br>2. Finish the line by **Finish by**. **Read time** is the length of the reference read in the narrated version.<br>3. Record each scene as its own take. Scene windows: 0:00–0:19, 0:19–0:38.0, 0:38.0–1:04.7, 1:04.7–1:46.2, then the title card 1:46.2–1:56 (no narration).<br>4. The boom lands at 1:29.9. Read the conclusion (4.7 to 4.10) after it, with a full pause before "Distance…", then stay silent through the title card to 1:56.<br>5. Pace is about 2.2 words per second, with a short breath at each comma. | **MCS**: "M-C-S", letter by letter<br>**Dijkstra's**: DIKE-struhz<br>**MAP-A**: "Map A"<br>**Albay**: al-BYE<br>**jeepney**: JEEP-nee<br>**Goal 11**: "Goal eleven"<br>**e-mail**: EE-mail<br>**"test network"**, never "Albay", for the a3, S3 and S4 numbers<br>**RA 10932**, ₱ fares and table values are shown on screen only. Do not read them. |

### Scene 1 · Trip

0:00.0–0:19.0 · 41 words · bar: Convenience

| # | Start | Finish by | Read time | Narration (read this) · on screen |
|---|---|---|---|---|
| 1.1 | 0:00.3 | 0:03.3 | 2.9 s | **Start with one trip, from point A to point B.**<br>On screen: Albay map draws in; pins A and B drop; a dotted path with a "?" joins them. |
| 1.2 | 0:03.8 | 0:07.5 | 3.7 s | **Which jeepney do you ride, when do you arrive, and what do you pay?**<br>On screen: Three tags pop in, one per question: Which jeepney? · When do you arrive? · What do you pay? |
| 1.3 | 0:08.1 | 0:13.3 | 5.1 s | **In Albay, no machine-readable record of routes, stops, or fares can tell you.**<br>On screen: The map dims while unrecorded vehicles drive between towns. On "no machine-readable record" the tags turn grey; the notation "R = S = H = F = empty set" (routes · stops · headways · fares); chips Jeepney · Tricycle · Minibus · UV van and "No routes, stops, headways, or fares on record", with "fares" in gold. |
| 1.4 | 0:13.9 | 0:15.7 | 1.7 s | **Our timetable model can.**<br>On screen: Synthetic test network with "Timetable graph G = (V, E)"; gold route a3 → S4; card "Door to door: 18.6 min" with τ(a3, S4) = 18.6 min; the fare meter counts to ₱20.00 for 7 km on a traditional jeepney, then "₱16.00 with 20% discount" and f(d) = 14 + 2·max(0, d − 4). Hold to 0:19. |

## Scene 2 · Community

0:19.0–0:38.0 · 42 words · bar: Fairness

| # | Start | Finish by | Read time | Narration (read this) · on screen |
|---|---|---|---|---|
| 2.1 | 0:19.2 | 0:20.2 | 0.9 s | **Zoom out.**<br>On screen: The camera pulls back from the route to Albay: town boundaries, main roads and the four urban centres. |
| 2.2 | 0:20.7 | 0:28.1 | 7.4 s | **The same model counts the schools, markets, and hospitals a person can reach in thirty minutes, and who is left out.**<br>On screen: Schools, markets and hospitals appear clustered around the town centres; $A_i(30) = \Sigma\ 1[\tau(i, j) \le 30\ \text{min}]$; a road-shaped 30-minute reach area draws around Legazpi City; facilities inside light up, those outside grey out; "$L = \{ i : A_i(30) = 0 \}$" on "who is left out". |
| 2.3 | 0:28.7 | 0:33.7 | 4.9 s | **Sustainable Development Goal 11 asks for transport accessible to all.**<br>On screen: Lower third: "SDG 11.2 · transport accessible to all". |
| 2.4 | 0:34.2 | 0:37.2 | 2.9 s | **In Albay, there was nothing to measure that against.**<br>On screen: A "?" appears on the 30-minute reach area. |

## Scene 3 · Patient

0:38.0–1:04.7 · 59 words · bar: A life

| # | Start | Finish by | Read time | Narration (read this) · on screen |
|---|---|---|---|---|
| 3.1 | 0:38.3 | 0:39.8 | 1.4 s | **Now, an emergency.**<br>On screen: The Albay map zooms back into the test network; the patient appears at a3 with a red pulse. |
| 3.2 | 0:40.4 | 0:44.5 | 4.0 s | **The nearest hospital may be full, or lack the right level of care.**<br>On screen: A red dashed line marks the nearest hospital, S3 (7.0 min), with $h_dist = \text{argmin}\ d(a3, h) = S3$. On "full" S3 turns red: $b(S3) = 0$; card S3 · FULL · Level 1 · 0 free beds · 7.0 min. |
| 3.3 | 0:45.2 | 0:50.0 | 4.8 s | **Maps show where hospitals are, not free beds or what a family can afford.**<br>On screen: Card "A map shows where. It does not show free beds, level, or cost." S4 shows "free beds ? · level ? · cost ?". |
| 3.4 | 0:50.8 | 0:59.2 | 8.3 s | **Our hospital mode collects free-bed reports by e-mail** [0:54.9] **and sends the ambulance to the nearest hospital that can take the patient.**<br>On screen: "HOSPITAL MODE"; six e-mails fly to S4 and the counter reaches 6/6; S4 turns gold, $b(S4) = 8$ (Level 3 · 18.6 min); the decision rule $h^* = \text{argmin}\ \tau(a3, h)$ subject to level of care, open ER, free beds and budget or emergency acuity; the ambulance carries the patient past S3; $h^* = S4$; check chips. |
| 3.5 | 1:00.0 | 1:03.3 | 3.2 s | **In an emergency, payment never delays the route.**<br>On screen: Chip "Payment waived for acuity 1 and 2 (RA 10932)" with "k in $K_E = \{1, 2\}$". After the line, hold: "Distance-only routing sent 5 of 6 anchors to a full hospital" (to 1:07). |

## Scene 4 · How MCS comes into play

1:04.7–1:46.2 · 101 words · bar: MCS

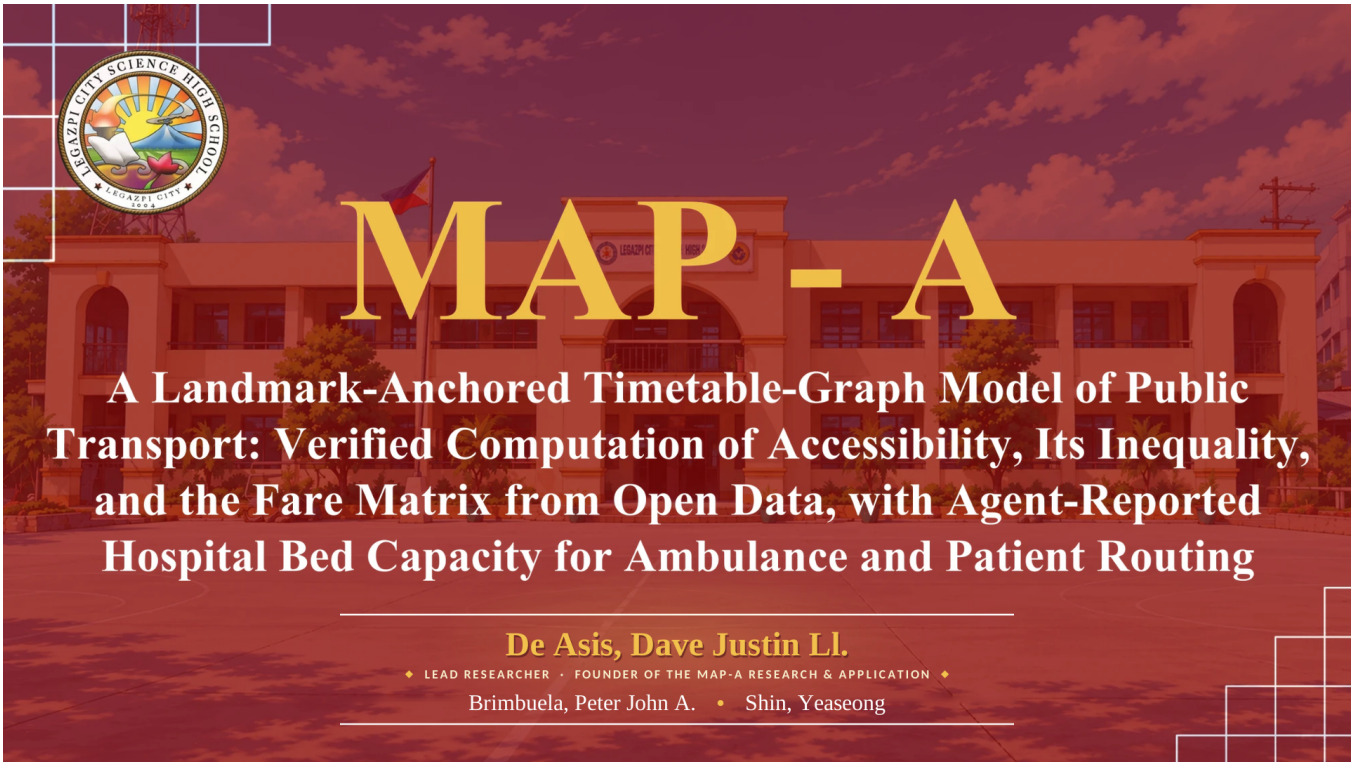
| # | Start | Finish by | Read time | Narration (read this) · on screen |
|---|---|---|---|---|
| 4.1 | 1:04.8 | 1:08.0 | 3.1 s | **You may be wondering: how does MCS come into play?**<br>On screen: Scene title. “You may be wondering…”, then “How does”. On “MCS” the gold letters M, C, S drop in and unfold into Mathematical · Computational · Sciences, then “come into play?”. Faint math symbols drift behind; the stakes bar climbs to its top rung, “MCS”. |
| 4.2 | 1:08.5 | 1:11.2 | 2.6 s | **Mathematics turns open data into a graph.**<br>On screen: The title shrinks into an “MCS” badge and a chain lights up across the top: Open data → Graph → Algorithm → Numbers → Optimisation → Proof. The synthetic test network builds stop by stop (a3, S3, S4). Panel “MATHEMATICS”: G = (V, E); V = stops and landmarks, E = ride, wait, transfer and walk arcs; built from open maps, registries, census counts and fares. |
| 4.3 | 1:11.6 | 1:16.3 | 4.6 s | **Computing searches it with Dijkstra's algorithm for the exact fastest trip.**<br>On screen: Panel “COMPUTING”: Dijkstra’s label-setting search, $\tau^*(o, d, t0) = t0 + \min$ over paths of $\Sigma\ \omega(u, v)$, with $\omega(u, v) = t(v) - t(u) \geq 0$. On “Dijkstra’s” a wave spreads from a3 and every stop settles with its travel time in minutes (S3 7.0 min, S4 18.6 min). On “fastest” the route a3 → S4 lights gold; card “Proposition 1 · returns the exact earliest arrival” and $\tau$(a3, S4) = 18.6 min. |
| 4.4 | 1:16.8 | 1:21.1 | 4.3 s | **Those times count access, measure inequality, and price every fare.**<br>On screen: On “access”: stops within 30 minutes of a3 glow and the rest grey out; panel $A_i(30) = \Sigma\ 1[\tau(i, j) \leq 30\ \text{min}]$. On “inequality”: a Lorenz curve and the population-weighted Gini coefficient $G = (1/2\mu)\ \Sigma\ \Sigma\ p_a\ p_b\ |A_a - A_b|$. On “fare”: a fare matrix fills cell by cell, $F_{ij} = f(d_{ij})$ with $f(d) = 14 + 2 \cdot \max(0, d - 4)$; the 7 km pair shows ₱20. |
| 4.5 | 1:21.6 | 1:26.5 | 4.9 s | **For a patient, it finds the fastest hospital with the right care and a free bed.**<br>On screen: Panel “OPTIMISATION”: $h^* = \text{argmin}\ \tau(a3, h)$ subject to $\ell(h) \geq \ell_k$ (right level of care), ER(h) = 1 (ER open), b(h) > 0 (a free bed), c(h) ≤ B or k  K_E (within budget, or an emergency). On “patient” S3 turns red: FULL · 0 free beds, “S3 is nearest, but b(S3) = 0”. The ambulance drives a3 → S4, each condition gets a check, and $h^*$ = S4. |
| 4.6 | 1:27.0 | 1:29.8 | 2.7 s | **Twelve results, proven, then verified.**<br>On screen: “12 results = 9 propositions + 3 corollaries”: twelve tiles (Props. 1, 2, 3, 5, 6, 7, 9, 10, 12; Cors. 4, 8, 11) get a proof mark on “proven” and a check on “then verified”, with “Each proven on paper, then checked against independent code on synthetic instances · no field data”. They implode into a glowing core as a riser builds. |
| 4.7 | 1:30.4 | 1:32.6 | 2.2 s | **MAP-A will not add one bed to Albay.**<br>On screen: Conclusion. A boom at 1:29.9 clears the screen to a plain background. The line appears large, with “one bed” in gold. The large text is its subtitle. |
| 4.8 | 1:33.1 | 1:38.3 | 5.3 s | **But once hospitals report, it will tell him which one has a free bed and the right care.**<br>On screen: Appears under it in two rows, with “free bed” and “right care” in gold. Both lines then fade out. |

| # | Start | Finish by | Read time | Narration (read this) · on screen |
|---|---|---|---|---|
| 4.9 | 1:39.4 | 1:41.9 | 2.4 s | [pause] **Distance tells you where the hospital is.**<br>On screen: [pause] The screen is empty for a moment. Then the line appears, with the MAP-A logo and title below. |
| 4.10 | 1:42.3 | 1:45.0 | 2.7 s | **Access tells you whether it can help in time.**<br>On screen: Lands in large gold type with a final hit; “in time” is underlined. Hold silence. At 1:46.2 the title card follows. |

## Title card · Researchers

1:46.2–1:56.2 · no narration

| # | Start | Finish by | Read time | Narration (read this) · on screen |
|---|---|---|---|---|
| 5.1 | 1:46.2 | 1:56 | — | [No narration. Stay silent; let the music and the title card play.]<br>On screen: 1:46.2 The conclusion frame crossfades into the title slide: school building under a red overlay, Legazpi City Science High School seal, white corner grid. 1:47.9 The gold “MAP - A” drops in letter by letter and lands with a hit; a gold shine sweeps across it. 1:48.2 The full title fades in line by line: “A Landmark-Anchored Timetable-Graph Model of Public Transport: Verified Computation of Accessibility, Its Inequality, and the Fare Matrix from Open Data, with Agent-Reported Hospital Bed Capacity for Ambulance and Patient Routing”. 1:49.5 Two white lines draw outward. 1:50.0 **De Asis, Dave Justin Ll.** in large gold type; 1:50.6 under it “LEAD RESEARCHER · FOUNDER OF THE MAP-A RESEARCH & APPLICATION”; 1:51.1 below, “Brimbuela, Peter John A. · Shin, Yeaseong” in white. 1:51.9 A gold shine and glow pass over the leader’s name. 1:55.5 Fade to black; the video ends at 1:56. |




Title card, final frame (also supplied as MAP-A_Title_Page_1920x1080.png and MAP-A_Title_Page_3840x2160.png for the title slide).

# Clean read

The same narration without the production notes. Use it as a teleprompter or to rehearse. A slash marks a short breath.

**SCENE 1 · TRIP · 0:00.0–0:19.0**

0:00.3 Start with one trip, / from point A to point B.

0:03.8 Which jeepney do you ride, / when do you arrive, / and what do you pay?

0:08.1 In Albay, / no machine-readable record of routes, / stops, / or fares can tell you.

0:13.9 Our timetable model can.

**SCENE 2 · COMMUNITY · 0:19.0–0:38.0**

0:19.2 Zoom out.

0:20.7 The same model counts the schools, / markets, / and hospitals a person can reach in thirty minutes, / and who is left out.

0:28.7 Sustainable Development Goal 11 asks for transport accessible to all.

0:34.2 In Albay, / there was nothing to measure that against.

**SCENE 3 · PATIENT · 0:38.0–1:04.7**

0:38.3 Now, / an emergency.

0:40.4 The nearest hospital may be full, / or lack the right level of care.

0:45.2 Maps show where hospitals are, / not free beds or what a family can afford.

0:50.8 Our hospital mode collects free-bed reports by e-mail and sends the ambulance to the nearest hospital that can take the patient.

1:00.0 In an emergency, / payment never delays the route.

**SCENE 4 · HOW MCS COMES INTO PLAY · 1:04.7–1:46.2**

1:04.8 You may be wondering: / how does MCS come into play?

1:08.5 Mathematics turns open data into a graph.

1:11.6 Computing searches it with Dijkstra's algorithm for the exact fastest trip.

1:16.8 Those times count access, / measure inequality, / and price every fare.

1:21.6 For a patient, / it finds the fastest hospital with the right care and a free bed.

1:27.0 Twelve results, / proven, / then verified.

1:30.4 **MAP-A will not add one bed to Albay.**

1:33.1 **But once hospitals report, it will tell him which one has a free bed and the right care.**

[pause]

1:39.4 **Distance tells you where the hospital is.**

1:42.3 **Access tells you whether it can help in time.**

**TITLE CARD · RESEARCHERS · 1:46.2–1:56.2**

1:46.2 [Silent. Title card: MAP - A · De Asis, Dave Justin Ll., Lead Researcher and Founder · Brimbuela, Peter John A. · Shin, Yeaseong. Ends 1:56.]